INVESTIGATIONS OF THE ENVIRONMENTAL IMPACT OF THE LADYSMITH SEWAGE TREATMENT PLANT ON LADYSMITH HARBOUR

By

Debbie Hansen

Ministry of Environment, Lands and Parks
Environmental Protection
Nanaimo, B.C.

September 1993

in association with
University of Victoria
Biology Co-op Program

ABSTRACT

Ladysmith Harbour, supporting a moderate amount of industrial, commercial, and recreational activities, is potentially affected in terms of water quality from a number of sources. One such source is the Ladysmith sewage treatment plant (STP). Presently, the STPs' effluent is meeting permit levels for B.O.D. and T.S.S. However, rainbow trout bioassays indicate the effluent to be toxic. Ammonia levels, and B.O.D. are likely key factors contributing to the effluent toxicity. Dissolved oxygen levels are depressed in the lower depths of the Harbour. Macoma bioassays indicated toxicity in the sediment, with neither of the sample sets surviving the test. Continued monitoring should be performed through out the year. Improving the water quality of Ladysmith Harbour and removing the shellfish closures currently in effect are of prime concern.

TABLE OF CONTENTS

	Page
ABSTRACT	i
TABLE OF CONTENTS	ii
LIST OF FIGURES AND TABLES	iii
1. INTRODUCTION	1
2. SITE DESCRIPTION	3
2.1 Oceanography	3
2.2 Fisheries Resource	5
2.2.1 Salmonids	5
2.2.2 Other Fish Species	5
2.2.3 Shellfish Resources	5
2.3 Sewage Treatment Plant Background	8
3. MATERIALS AND METHODS	10
4. EFFLUENT CHARACTERISTICS	13
4.1 Water Quality	13
4.2 Toxicity of Effluent	15
5. RECEIVING ENVIRONMENT	15
5.1 Marine Water	16
5.1.1 Physical Oceanography	16
5.1.2 Dissolved Oxygen	16
5.1.3 Nitrogen	17
5.1.4 Bacteriological	19
5.2 Marine Sediment	19
6. CONCLUSIONS	22
7. REFERENCES	24
APPENDIX I	26
APPENDIX II	29
PHOTOS	34

LIST OF FIGURES AND TABLES

		Page
Figure 1.	Ladysmith Harbour water quality monitoring study area	2
Figure 2.	Map showing oceanography of Ladysmith Harbour	4
Figure 3.	Map showing the location and type of each sampling site in Ladysmith Harbour	11
Table 1.	Fish escapement for the five creeks entering Ladysmith Harbour (1976-85)	6
Table 2.	Ladysmith sewage treatment plant permit PE-120	9
Table 3.	Water quality tests performed at each station in Ladysmith Harbour	12
Table 4.	Ladysmith sewage treatment plants' final effluent characteristics and acute toxicity	14
Table 5.	Water quality parameters for sites 4 and 5 in Ladysmith Harbour	18
Table 6.	Bacteriological data for six sites in Ladysmith Harbour	20
Table 7.	Summary of macoma bioassay results	21

1. INTRODUCTION

The Town of Ladysmith lies within the Cowichan Valley Regional District and is located along the Trans Canada Highway adjacent to Ladysmith Harbour. Nanaimo is to the north and Duncan is on the south. Figure 1 shows the town and treatment plant site within the region.

Ladysmith supports a population of approximately 5000. The economy of Ladysmith has been dominated primarily by industrial, commercial and recreational development.

The sewage treatment plant (STP) is located at Holland Bank on the west shore of Ladysmith harbour. Shoreline homes lie to the north and several single residences exist to the south. Ladysmith is located to the northwest, behind the plant site.

The STP provides primary treatment through screening, grit removal and collection of settleable solids prior to disinfection by chlorination and subsequent dechlorination. Screenings and grit are removed and disposed by landfilling. The sludge is pasteurized, stabilized and dewatered, and is suitable for agricultural use. The main building houses the headworks, autothermophilic aerobic sludge digesters and dewatering process. The sewage is then transported via pipe to the primary settling tank, sludge pumping and chlorine contact tank. Two outfalls combine to convey the final effluent and storm water overflow to Ladysmith Harbour to a point approximately 810 m offshore at a discharge depth of about 20 m.

In the past 10 years, little work has been performed on the water quality of Ladysmith harbour. Annual testing of the effluent is completed in regards to the Ladysmith STP permit PE-120 (Appendix I). Therefore, the present effects of the discharge upon important resources such as salmonids and shellfish need to be updated. This report presents information from two surveys done through the summer of 1993 to gain knowledge on the immediate or long term impacts, if any, of this effluent on Ladysmith Harbour. Toxicity and water chemistry were examined on both the discharge and the receiving environment at 6 sites in Ladysmith Harbour.

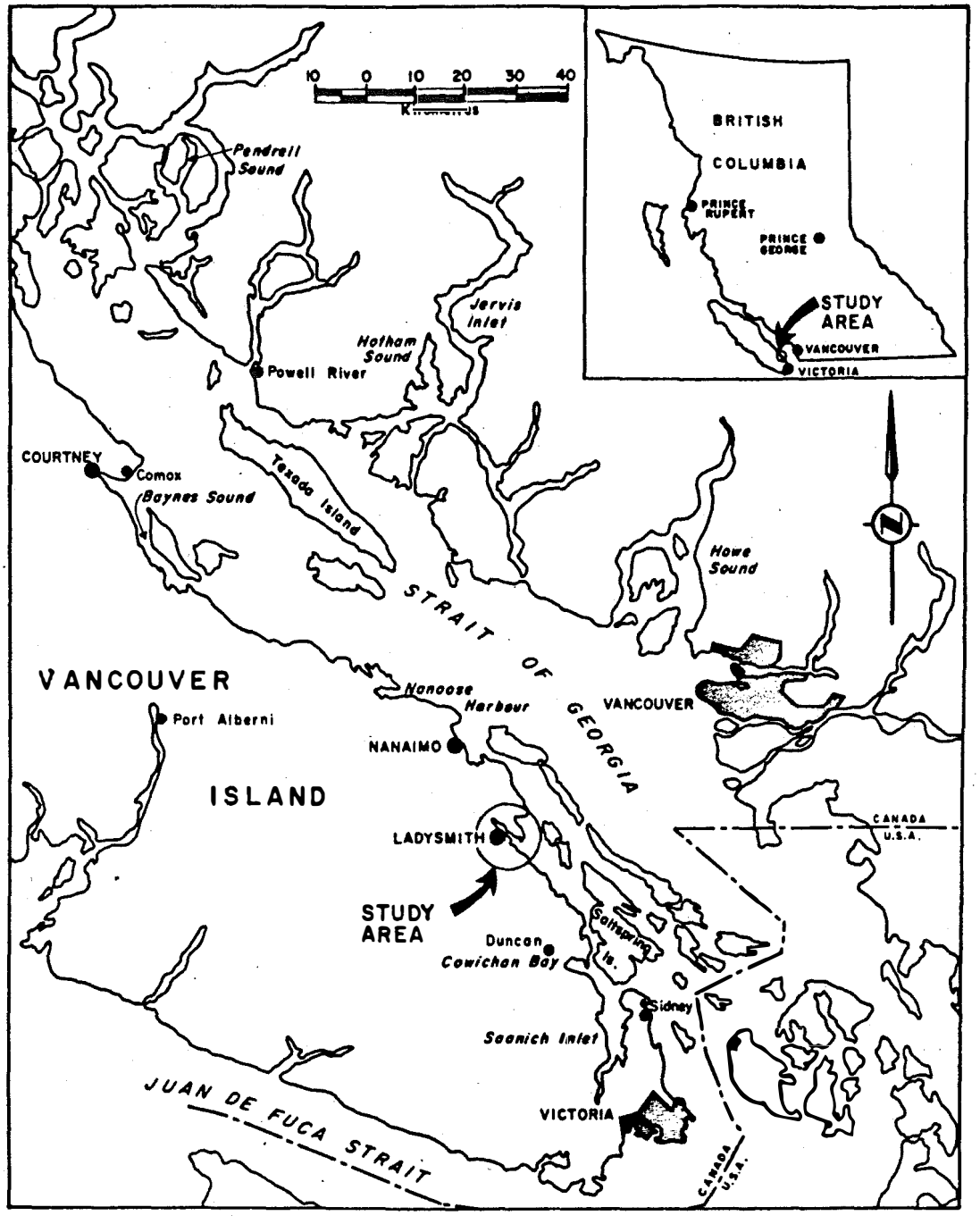


FIGURE 1: Ladysmith Harbour Water Quality Monitoring Study Area

2. SITE DESCRIPTION AND HISTORY

2.1 OCEANOGRAPHY

Ladysmith Harbour is situated on the southeast coast of Vancouver Island (Fig. 2). It is a small inlet which is oriented in a northwest-southeast direction and has a length of 9 km, shoreline length of approximately 25 km and surface area of approximately 1120 ha. The harbour is split into inner and outer regions via Slag Point, a constriction of the western shoreline which extends eastward toward the southern tip of Woods Islands. Stag Point offers protection to the inner harbour while exerting only minor influences on tidal flushing and circulation (Anon., 1981a).

The inner harbour is characterized by its extensive intertidal flatlands, shallow water of less than 10 m and greater variations in water temperature than the outer harbour (McDougall and Boyd, 1984). Four creeks drain into the inner harbour causing variations in surface salinity and water quality depending on the quantity of freshwater discharged. There are two sawmills situated on the west shore of the inner harbour. The harbour provides protection for both intertidal and subtidal log storage.

The outer harbour increases in depth to approximately 50 m and has a larger volume than the inner harbour. As only one creek drains into the outer harbour, Stuart Channel plays a greater role on the physical oceanography and to a lesser degree water from the inner harbour.

McDougall and Boyd (1984) discuss a two-layer system of circulation within Ladysmith Harbour due to the presence of freshwater. Outflows tend to occur in the upper layers with inflows occurring in the lower layers for most of the year. As low density surface water flows out of the harbour, it entrains more saline water from below developing a slow compensatory inflow.

Flushing of the inner harbour area is considered to be poor, since it is dependent almost exclusively on the exchanging of water by the tide with some assistance from wind mixing (Anon, 1981a). Pommen and Holden (1981) have shown that surface drogues deployed at the Ladysmith Harbour STP outfall plume can move into the inner harbour. This movement is accelerated by SE winds, which tend to blow surfaced effluent towards the head of the harbour. Thus although the head of the harbour will experience almost 100% water exchange during a tidal cycle, much of the same water may return and re-enter the inner harbour.

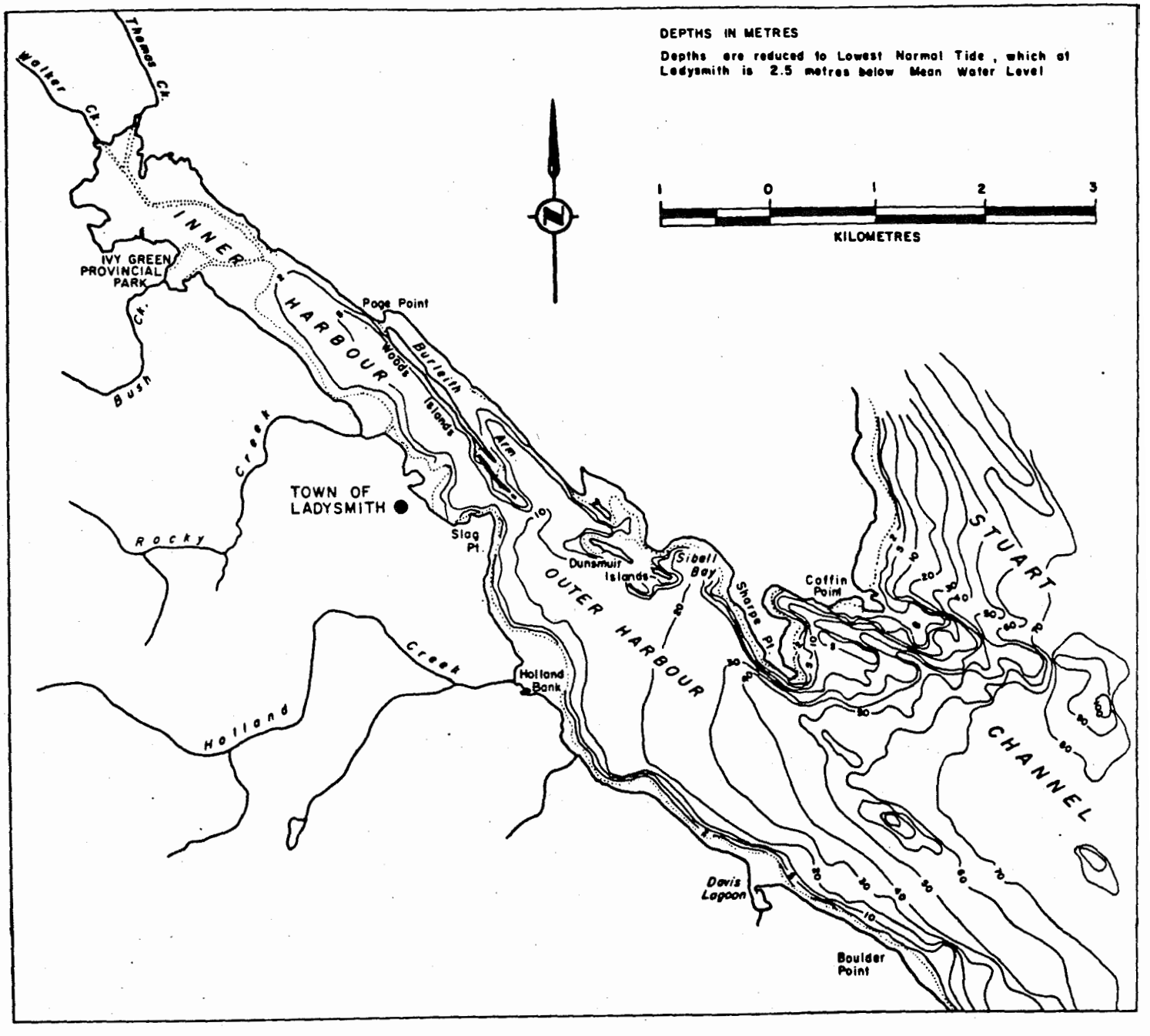


FIGURE 2: Ladysmith Harbour

2.2 FISHERIES RESOURCE

2.2.1 Salmonids

There are five creeks which flow into Ladysmith Harbour: Bush Creek, Holland Creek, Rocky (Tyee) Creek, Thomas Creek and Walker Creek (Fig. 2). Chum (Oncorhynchus keta) and coho (O. kisutch) salmon utilize all streams flowing into the harbour. Cutthroat (Salmo clarki) trout are present in all creeks but one, Bush Creek. Steelhead (S. gairdneri) have been observed in Holland Creek. Table 1 indicates the historical escapements for all five creeks (Department of Fisheries and Oceans, 1988).

Restricted accessible area has extremely limited the spawning grounds to the lower reaches in each of the five creeks. While spawning habitat may be sparse, potential nursery areas for juvenile salmonids, such as eelgrass and marsh grass beds are located along the inlets' shoreline. Salt marsh communities exist in deltas at the mouth of Bush Creek (140 acres) and Holland Creek (60 acres). Tidal drift of insects from the marshes along with copepods and isopods from the lower mud flats constitute one of the principal sources of food for rearing salmonids (Patterson, 1975).

2.2.2 Other Fish Species

In the 1950's and 60's Ladysmith Harbour was an ideal location for herring spawn. A survey by Patterson (1975) revealed that there has been a reduction in the areas utilized by herring as spawning grounds to only Dunsmuir Island, Sibell Bay and Sharpe Point. Reduction of eelgrass, rockweed and japweed beds within the harbour may be one of the major contributing factors in the failure of herring to spawn within Ladysmith Harbour.

Personal communication with the DFO offices in Nanaimo revealed that herring spawn does occur in small numbers, but is limited to the outer regions of the harbour (ie. Coffin Point).

2.2.3 Shellfish Resources

Oysters

The marine environment of Ladysmith Harbour offered favourable conditions for oyster production for many years. However, despite successful oyster breeding in harbour waters, bacteriological contamination of oyster growing waters caused subsequent closure of most of the harbour to shellfish harvesting in 1965 (McDougal and Boyd, 1984). Re-opening of commercial

Table 1. Fish escapement for the five creeks entering Ladysmith Harbour (1976-85).

Creek	Coho		Chum		Cutthroat		Steelhead	
	Mean	Max	Mean	Max	Mean	Max	Mean	Max
Bush	128	290	2878	10500	-	-	-	-
Holland	27	60	3807	10500	present		present	
Rocky (Tyee)	4	15	3	10	present		-	-
Thomas	present		-	-	present		-	-
Walker	110	250	770	1950	present		-	-

oyster leases in the inner harbour in 1975 also showed a decline in oyster production. Water quality deterioration was the suspected cause (Bourne et al., 1981).

In 1979, a 48-hour oyster bioassay used to assess water quality of sea water samples from Ladysmith Harbour indicated that developing oyster embryos were seriously affected by some form of contamination in the waters (Bourne et al, 1981) As a consequence, the Ladysmith Harbour Crown Foreshore Plan (Anon., 1981a) recommended a comprehensive water quality monitoring program to investigate coliform bacteria and chlorophenol contamination in the harbour.

In the summer of 1982, the same 48-hour oyster embryo bioassy used in 1979 was performed on potential contaminants. Results indicated improved quality of the inner harbour waters, however diluted sewage effluent showed poorer results. In their report, McDougall and Boyd (1984) concluded the water quality in Ladysmith Harbour was suitable for successful oyster culture and spawning, with the sewage discharge exerting only a minor impact on the receiving water quality in the outer harbour in terms of oyster culture.

During 1983 and 1984, bacteriological and sanitary surveys were conducted by the Environmental Protection Service in Ladysmith Harbour. Results indicated that shellfish growing water bacteriological standards were exceeded in portions of the shellfish harvesting area in the open area of the inner harbour during periods of extreme rainfall (Kay and Walker, 1985). The closed shellfish harvesting areas of the outer harbour (ie. Sibell Bay and Dunsmiur Island) revealed no contaminants during any of the studies.

Presently there is a closure on all oyster harvesting in Ladysmith Harbour from Sharpe Point to Boulder Point. If the closure was lifted there are 11 oyster leases present in Ladysmith Harbour, each having the opportunity to harvest.

Other Shellfish

Data on the value of other shellfish resources is very limited. There is no commercial fishery within the study area boundaries. It is known, however, that the harbour supports an extremely small sports fishery for crabs. The harbour is too shallow for a shrimp fishery. A sports fishery for clams did exist on the extensive tidal mud flats at the head of the harbour. However, due to shellfish closures, no harvesting currently takes place.

2.3 STP BACKGROUND

Prior to 1965 an estimated 1,575,000 L of raw sewage was discharged daily into the inner harbour via a short outfall pipe located on north side of Slag Point (Tevendale, 1973a). With the construction of the primary STP, sewage discharge was diverted from the inner harbour to the outer harbour while the inner harbour outfall was converted to handle storm sewer overflows.

Several surveys in the early 1970's indicated that sewage contamination from local sources, ie. some sewage lines were still connected to the storm sewer overflow system and faulty septic fields, continued to affect much of the inner harbour. The surveys also revealed deterioration of the water quality in the outer harbour as a result of diverting most of the town's sewage to Holland Bank for primary treatment and offshore discharge.

In 1974, the STP was upgraded with flow-metering and chlorination. In addition, faulty septic tank absorption fields were upgraded in the inner harbour area.

A 1975 survey (Cooper and Kay, 1975) found improvements in water quality in both the inner and outer harbour. However contamination was still occurring.

By 1982 the storm water overflow had been diverted to the outer harbour and installation of the dechlorination system at the STP was in effect.

A report by Kay and Walker (1985) indicated that discharges of untreated combined storm and sanitary effluent by way of the newly constructed bypass outfall into the outer harbour did not exert a negative impact on the water quality in the Sibell Bay and Dunsmuir Island area. However the discharge of the sludge through the STP outfall was found to have a measurable effect on water quality in the Sibell Bay area.

In 1988 autothermophilic aerobic sludge processing facilities were added. This was to stop further sludge discharge into Ladysmith Harbour.

Table 2 indicates the amended and present permit PE-120 requirements for the effluent quality at the plant. Overflow patterns for past years indicated low to nil values during the dry months (May - October) and excessive values during the wet months (November-April). These overflow effluent disharges exceed the permitted values on daily and weekly periods during wet weather. Dayton and Knight (1991) proposed two options dealing with these overflow values and the expansion of the Ladysmith township itself. One consideration is to upgrade the

Table 2. Ladysmith sewage treatment plant permit PE-120.

Date	Permit Regulations	
Prior to 1988	Maximum rate through plant	3,200 cu m/d
	Maximum rate for bypass outfall	54,000 cu m/d
	T.S.S. 130 mg/L	
	B.O.D. 130 mg/L	
Amended 1988	Maximum rate through plant	6,100 cu m/d
	Annual average rate	3,050 cu m/d
	Maximum rate for bypass outfall	54,000 cu m/d
	T.S.S. 130 mg/L	
	B.O.D. 130 mg/L	
Amended 1993	Maximum rate through plant	6,100 cu m/d
	Annual average rate	3,050 cu m/d
	Maximum rate for bypass outfall	54,000 cu m/d
	Annual maximum rate	227,272 cu m/yr
	T.S.S. 130 mg/L	
	B.O.D. 130 mg/L	

Note: T.S.S. = Total Suspended Solids
B.O.D. = Biochemical Oxygen Demand

primary plant and extend the outfall beyond the embayment line to open marine waters. This would require a 3000 m extension of the existing outfall. The other option is to keep the existing outfall and convert the primary plant to a secondary plant.

Conversion of the plant to secondary treatment would require B.O.D. and T.S.S. values to be lowered to 45 and 60 mg/L respectively. Reduced B.O.D. levels would correlate with a decrease in the amount of organic materials present in the effluent. Reduction in ammonia levels through nitrification would also be observed.

3. MATERIALS AND METHODS

Samples were collected from a number of locations in and around Ladysmith Harbour on May 20 and July 6, 1993. The tides and time of sampling are as follows: May 20 from 1100 to 1255 hrs (pst) with a flood tide low at 1140 hrs; July 6 from 0840 to 1055 hrs (pst) with an ebb tide high at 0655 hrs. Figure 3 indicates the position of each site and nature of the sampling.

Marine water samples were collected at just below the surface at all sites by immersing the appropriate collection bottle. Other samples were taken at various depths for each station by using a Van Dorn water collection bottle. All samples were subsequently shipped in coolers to Zenon Environmental in Burnaby for analysis, as outlined in Clark and Shera (1985). A summary of the water quality tests for each of the stations is listed in Table 3.

At each of the marine stations, a variety of parameters were measured through out the water column using a Hydro Lab Surveyor II. These included temperature, dissolved oxygen (DO), pH, oxidation reduction potential (ORP) and salinity. Results of these physical oceanographic measurements are presented in Appendix II.

Bottom sediments were obtained using an Eckman grab. Numerous grabs (30-40) were made at each site to obtain sufficient material for analysis (approx. 1/4 to 1/3 cu ft). The sediment samples were shipped to the Aquatic Toxicity Lab for Macoma and Microtox bioassays. Subsamples of approximately 500 ml were sent to Zenon Labs for a modified SWEP (salt water extraction process) analysis, upon which an Echinoderm fertilization bioassay was performed.

The Ladysmith sewage treatment plant was sampled at the final access point to the effluent, just beyond the dechlorination line. Beyond this point, the effluent drops down into the

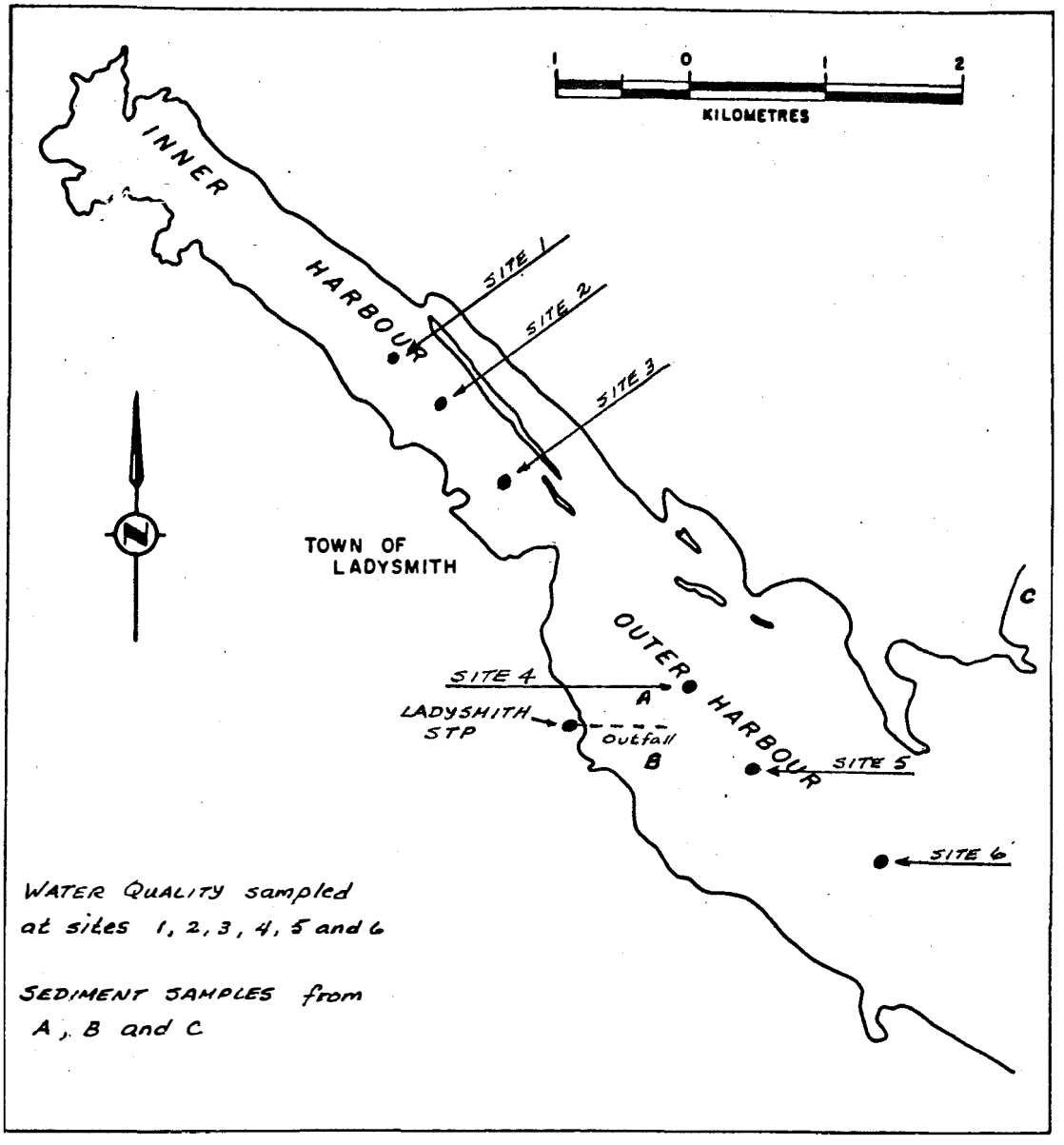


Figure 3. Location and type of sampling site in Ladysmith Harbour.

Table 3. Water quality tests performed at each station in Ladysmith Harbour.

Location	Depth (m)	Fecal coliform	Entero-cocci	Ammonia	Nitrate/ Nitrite	N-T	P-T
LH #1	0.5	X	X				
	5.0	X	X				
LH #2	0.5	X	X				
	5.0	X	X				
LH #3	0.5	X	X				
	9.0	X	X				
LH #4	0.5	X	X	X	X	X	X
	10.0	X	X	X	X	X	X
LH #5	0.5	X	X	X	X	X	X
	10.0	X	X	X	X	X	X
	18.0	X	X	X	X	X	X
LH #6	0.5	X	X				

discharge pipe and is not accessible.

Bioassays were carried out by the Aquatic Toxicity Lab as per guidelines specified in Provincial Guidelines and Laboratory Procedues for Measuring Acute Lethal Toxicity of Effluents to Fish (B.C. Bioassay Task Force, 1982).

Effluent samples were collected on May 20 and July 8, 1993 from the end of the chlorine contact chamber at the Ladysmith STP. Two 20 litre collapsable containers and six 20 litre collapsable containers were filled for the respective dates.

The effluent was sent to Aquatic Toxicity Lab where rainbow trout and chinook bioassays were performed. As well as a Microtox bioassay.

Additional water samples were collected for analysis of biochemical oxygen demand (B.O.D.), fecal coliforms, enterococci, nitrogen (ammonia, nitrate/nitrite, organic and total), total phosphorus, total suspended solids (T.S.S.) and total metals. These samples were sent to Zenon.

4. EFFLUENT CHARACTERISTICS

4.1 Water Quality

The final effluent was sampled twice during the summer of 1993 (May 20 and July 8). As shown in Table 4, the B.O.D. and T.S.S. were well within their permit limit of 130 mg/L with B.O.D. values of 67 and 70 mg/l and T.S.S. values of 30 and 17 mg/L for each respective sampling date. B.O.D. values were always relatively higher than T.S.S. values. Previous data collected yearly for the STP effluent indicate B.O.D. and T.S.S. values of under the 130 mg/L permit limit (Appendix I). Exceptions occur in October, 1988 and September, 1990 with B.O.D. values of 140 and 136 mg/L respectively.

The nitrogen data for the two sample dates differ. The initial sample had low ammonia (less than 0.005 mg/L and high organic nitrogen (21.6 mg/L) values. This could signify that ammonia is being broken down in the STP system, resulting in higher organic nitrogen values. The second set of results however were quite the opposite with high ammonia of 23.2 mg/L and lower organic nitrogen of 7.80 mg/L. This suggests that the ammonia is not properly being broken down. There are a number of possibilities which could lead to these values such as water flow levels, sludge removal, etc.. McDougall and Boyd (1984) suggest that lack of storm water dilution would likely result in higher mean values of ammonia. Two sets of sample data seems

Table 4. Ladysmith sewage treatment plant's final effluent characteristics and acute toxicity.

Parameter	May 20, 1993	July 8, 1993
B.O.D.	67.0 mg/L	70.0 mg/L
T.S.S.	30.0 mg/L	17.0 mg/L
Ammonia	<0.005 mg/L	23.2 mg/L
Nitrate/ Nitrite	0.12 mg/L	<0.02 mg/L
Organic Nitrogen	21.6 mg/L	7.80 mg/L
Total Nitrogen	21.7 mg/L	31.0 mg/L
Phosphorus	3.92 mg/L	4.00 mg/L
Fecal Coliforms	8300 CFU	3800 CFU
Enterococcus	2000 CFU	15 CFU
96hr LC50		
Rainbow Trout*	-	74.8 %
Chinook	-	<33.0 %
LT50 for rainbow trout at 100%	48 hrs	<24 hrs

* Note: A 96hr LC50 was performed on September 30, 1991 on rainbow trout yearlings with a result of 74.8%

insufficient to get an unbiased explanation of the results. However for comparison a September 1991 ammonia was recorded at 5.88 mg/L. This data suggests that ammonia levels are relatively unstable and range from virtually undetectable to at least 23 mg/L. Total nitrogen for each date was 21.7 and 31.0 mg/L respectively.

Phosphorus levels were almost identical for each date of sampling (3.92 and 4.00 mg/L). The phosphorus to nitrogen ratio ranged from 5.4 to 7.8 for the two sampling dates.

Bacteriological data for the effluent decreased from the initial sample on May 20. Fecal coliforms dropped from 8300 to 3800 CFU and enterococci decreased from 2000 to 15 CFU. These values range from moderate to low in respect to acceptable levels. Past data indicates fecal coliform values ranging anywhere from 100 to 2,000,000 CFU (Appendix I).

4.2 Toxicity of Effluent

A bioassay for acute rainbow trout toxicity (96hr LC50) was performed on the July 8th sample. The rainbow trout bioassay indicated an LC50 of 74.8%. A previous rainbow trout bioassay sample taken on September 30, 1991 had the same LC50 value of 74.8%. Even though both dates indicate the same LC50 value it is somewhat biased to compare them as different factors could have lead to the resulted value. For example B.O.D. levels for July 1993 and September 1991 were different at 70 and 116 mg/L respectively. Ammonia levels were relatively higher for July 1993 sample but alas unknown for September 1991. For comparative purposes, a chinook bioassay was run on the July 8th sample. This resulted in a 96hr LC50 value of less than 33%, obviously highly toxic. It appears that chinook were more sensitive to the final effluent than were rainbow trout.

For the May 20th sample, not enough effluent was collected to perform the 96hr LC50, however an LT50 (lethal time to kill 50% of the sample size) at 100% concentration was carried out on rainbow trout. This resulted in a value of 48 hrs. An LT50 value of less than 24 hrs was extrapolated from the July 8 data.

Overall it could be said that the effluent has remained relatively stable but somewhat toxic. The July toxicity is likely due to the presence of high ammonia. However, it should be noted that the practical difficulty of sampling the effluent further "downstream" of the dechlorination point, may have resulted in a poorly mixed sample and a high chlorine level. However, sample aeration prior to and during the bioassay procedure should rapidly dissipate the chlorine.

5. Receiving Environment

All pertinent receiving environment data for water chemistry is presented in Tables 5 and 6. Appendix II contains the hydrolab depth profiles for each of the six sites in the harbour.

5.1 Marine Water

5.1.1 Physical Oceanography

Temperatures for the May sample data ranged from 8.52 to 17.25 C while the July temperatures were slightly warmer ranging from 9.88 to 18.13 C (Appendix II). As to be expected warmer waters were found at the head and cooler waters at the mouth of the harbour. This condition apparently results from the large intertidal zone acting as a heat-sink during low tide periods, thus heat is subsequently transferred to overlying water during flood tides in the inner harbour (McDougal and Boyd, 1984). Outer harbour temperatures are lower due to a greater degree of mixing with colder water from Stuart Channel.

Salinities recorded in May ranged from 25.80 to 28.90 ppt while those in July ranged from 25.10 to 28.60 ppt (Appendix II). The second sampling data overall had lower salinity values. The reduced surface salinites suggest the freshwater influence of Thomas, Walker, Bush, Rocky and Holland Creeks (McDougall and Boyd, 1984).

5.1.2 Dissolved Oxygen

The maintenance of dissolved oxygen concentrations is essential for the protection of aquatic life. Significant stress can be placed upon organisms through the combination of low dissolved oxygen concentrations and toxic chemicals (Davis, 1975). The toxicity of some substances, such as ammonia, are enhanced by low dissolved oxygen. As well, higher water temperatures increase the adverse effects of low dissolved oxygen on fish.

Monitoring on May 20, 1993 indicated dissolved oxygen concentrations ranging from 7.08 to 14.00 mg/L (Appendix II). Most values were between 9.00 to 12.00 mg/L. By July 8, 1993 the dissolved oxygen tended to be approximately 3 to 4 mg/L lower, ranging from 4.66 to 10.43 mg/L. Most values were between 6 to 8 mg/L. Cooler water temperatures in May play a role in the higher DO values observed.

Dissolved oxygen concentrations generally decreased as the depth increased. Exceptions occurred around the 4 m level where an increase in DO was observed. This could be due to the presence of the two-layer circulation system found in Ladysmith Harbour where outflows occurr in the upper layers and inflows in

the lower layers. A phytoplankton maxima around that depth could also have lead to the increase in dissolved oxygen. Due to poor flushing of the inner harbour, DO values in July tended to be quite low.

The approved and working criteria for water quality recommends 6.5 mg/L of DO for aquatic marine life (Pommen, 1989). Presently only the upper portions of Ladysmith Harbours' water column meets this recommendation. As to how much the STP effluent contributes to the decrease in DO at lower depths is indeterminable. There are other factors in the harbour which play a role in lowering the DO. Logging and mud flats located in the inner harbour have some influence. During summer months circulation is poor, due to calm winds, and less freshwater is drained into the harbour. Collectively, these factors plus increased water temperature all work together to decrease the dissolved oxygen present in Ladysmith Harbour.

5.1.3 Nitrogen

Nitrate, nitrite and ammonia are the major inorganic nitrogen compounds occuring in surface waters. Two factors, pH and temperature of the water, are important to the toxicity of ammonia (Nordin and Pommen, 1986). Therefore in the receiving environment, the principle effects of ammonia input are: a) excessive ammonia levels can be toxic to aquatic life; b) nitrification of excessive ammonia input will utilize substantial portions of dissolved oxygen; c) input of ammonia may lead to nuisance phytoplankton blooms, which as part of their seasonal cycle may lead to depressed dissolved oxygen. Organic nitrogen in high levels will over time decompose, potentially leading to oxygen depletion.

At sites 4 and 5 located on either side of the STP discharge, the nitrogen data appears somewhat inconsistent (Table 5). This may be a reflection of the variation of ammonia and organic nitrogen results from the STP effluent. Organic nitrogen levels at site 4 remained relatively similar with only slight fluctuations (0.10 to 0.15 mg/L). However for site 5, organic nitrogen levels in July (0.04 mg/L) decreased by approximately 5 times those of May (0.20 mg/L). Ammonia levels at site 4 were higher for July data than May. At the surface in May ammonia was virtually undetectable (less than 0.005 mg/L), however in July surface levels were 0.019 mg/L. The site 5 ammonia levels varied as well. The May data was consistent with site 4 in that low levels were found at the surface (0.005 mg/L) and higher levels found at the bottom (0.099 mg/L). The July data had its' lowest ammonia value of less than 0.005 at the bottom (18.0 m), it then increased to 0.025 mg/L at 10.0 m and finally lowered to 0.016 mg/L at the surface. These results could be a reflection of the STP effluent data where in July the organic nitrogen level and ammonia level were opposite than the May data.

Table 5. Water quality parameters for sites 4 and 5 in Ladysmith Harbour (all measured in mg/L).

Ladysmith #4:

Parameter	May 20		July 8	
	0.5m	10.0m	0.5m	10.0m
Organic N	0.13	0.13	0.10	0.15
Total N	0.13	0.15	0.12	0.21
Ammonia	<0.005	0.021	0.019	0.027
NO3/NO2	<0.02	<0.02	<0.02	<0.03
Phosphorus	0.028	0.051	0.042	0.051

Ladysmith #5:

Parameter	May 20			July 8		
	0.5m	10.0m	18.0m	0.5m	10.0m	18.0m
Organic N	0.17	0.25	0.20	0.05	<0.04	0.04
Total N	0.17	0.27	0.35	<0.09	<0.06	0.17
Ammonia	0.005	0.021	0.099	0.016	0.025	<0.005
NO3/NO2	<0.02	<0.02	0.05	<0.02	<0.02	0.13
Phosphorus	0.023	0.045	0.061	0.042	0.046	0.076

Nitrate/nitrite levels were virtually undetectable (less than 0.02 mg/L) for both sites. The 18.0 m level at site 5 had slightly higher values at both sample dates of 0.05 and 0.13 mg/L respectively.

The provincial total ammonia criteria for the protection of aquatic life in marine waters are as follows: maximum concentration of 2.5 mg/L; average concentration of less than 1.0 mg/L (based on 5 samples in 30 days). At no time in any of the sampling done at Ladysmith Harbour were potentially toxic levels of ammonia identified. In fact, the highest ammonia concentration found was 0.099 mg/L. However ammonia can be a nutrient source for phytoplankton and utilize oxygen in nitrification. Both can have detrimental effects on the receiving environment.

5.1.4. Bacteriological

Fecal coliform and enterococci levels remained virtually undetectable (less than 1.0 CFU) in the majority of samples throughout the harbour (Table 6). Fecal coliforms, when detected, had a slightly higher value than enterococci values with a maximum value of 9.0 CFU. This remains below the fecal coliform value of 14 CFU selected for the protection of shellfish harvesting (Pommen, 1989).

5.2 Marine Sediment

Sediment samples were obtained on both dates, however the May sample was taken on a trial basis and limited to a Microtox bioassay only. The July sediment sample was expanded to include other bioassays. The sediment samples, taken from either side of the STP outfall, had a distinct hydrogen sulphide odour and were black in color. It should be noted that the comparison sediment taken from outside Ladysmith Harbour near Coffin Point also had a strong hydrogen sulphide odour.

Microtox is a marine strain of bioluminescence photo bacteria which emits a blue green light as a natural function. When challenged by a toxicant, a decrease in light output is measured and an EC50 (effective concentration causing 50% light decrease) is recorded. The Microtox bioassay results from Ladysmith Harbour A and B sites for both sample dates indicated no significant toxicity to the bacteria (greater than 1% EC50). The results ranged from a minimum value of 2.57% (Ladysmith Harbour B, May 20) to a maximum value of 10.95% (Ladysmith Harbour A, July 8). The control sample could not be computed due to the sediment particle size.

Macoma balthica, a clam common to B.C., is one of the few species that actually filters sediment particles. This makes it

Table 6. Bacteriological data for six sites in the Ladysmith Harbour.

Site	Date	Depth (m)	Fecal coliforms (CFU)	Enterococci (CFU)
E218911	930520	0.5	1.0	<2.0
		5.0	1.0	1.0
	930706	0.5	<1.0	<1.0
		5.0	<1.0	<1.0
E218912	930520	0.5	3.0	<2.0
		5.0	3.0	<2.0
	930706	0.5	1.0	<1.0
		5.0	<1.0	<1.0
E218913	930520	0.5	<2.0	<2.0
		9.0	2.0	<2.0
	930706	0.5	<1.0	<1.0
		9.0	9.0	1.0
E218914	930520	0.5	<2.0	<2.0
		10.0	<2.0	<2.0
	930706	0.5	<1.0	<1.0
		10.0	<1.0	<1.0
E218915	930520	0.5	<2.0	1.0
		10.0	6.0	<2.0
		18.0	4.0	2.0
	930706	0.5	<1.0	<1.0
		10.0	<1.0	1.0
		18.0	1.0	<1.0
E218916	930520	0.5	<2.0	<2.0
	930706	0.5	<1.0	1.0

Note: See Fig. for reference to sites.

Table 7. Summary of Macoma bioassay results.

Site	Dead	Alive	Missing	Surfaced	Comments
Ladysmith A	50	0	0	0	Strong H2S odour.
Ladysmith B	50	0	0	0	Strong H2S odour.
Ladysmith C* (Comparison)	12	38	0	0	Strong H2S odour.
Roberts Bank	4	50	0	0	All clams come from this site.

* Some of the shells had black edges.

ideal to use for toxicity testing. Macoma bioassay results indicate that both sediments A and B were toxic to clams over the 30 day period (Table 7). Observations throughout the test period noted that within the first two weeks 50% of the clams were dead and by 30 days all clams (50 per site) were deceased.

Unexpectantly the comparison sediment had 24% (12 dead) mortality and 36% (18) of the clams surfaced. Presence of the clams on the surface indicates they rejected the sediment, as their natural instinct is to bury themselves. It is believed that the strong presence of hydrogen sulphide and depleted oxygen levels in the sediment is the probable cause of mortality. The tolerance level of hydrogen sulphide for Macoma is 17.6 um per litre. It should be noted that the lab's sediment control from Roberts Bank had 4 dead clams.

The echinoderm fertilization bioassay first requires a modified SWEP to be performed on the sediment samples. This process uses salt water to extract any toxicants which happen to be in the sediment. Echinoderm gametes are then exposed to this extraction of various concentrations, incubated and mixed with eggs. After a period of incubation the eggs are studied to record percent fertilization. This bioassay is also ideal for determining receiving environmental impacts. The bioassay was conducted at 100% concentration of the modified SWEP. The control experiment resulted in 95% fertilization of the echinoderm gametes. Site B and site C showed similar results with 88% and 86% fertilization respectively, indicating minimal toxicity. Due to variations in the lab results for Site A, the test is currently being reconducted.

6. CONCLUSIONS

At present B.O.D. and T.S.S. values are well within their permitted limit for the Ladysmith STP effluent, however rainbow trout bioassays indicate that the effluent is acutely toxic. Of concern was the alternating high value of ammonia and organic nitrogen between the two sample dates.

Water quality in the harbour indicates that DO levels may be be affected. Dissolved oxygen concentrations are currently below the acceptable standards in the lower depths of the water column. However, in addition to the STP, there are a number of other factors which may contribute to depressed dissolved oxygen levels, such as log booming, inner harbour mud flats and other natural phenomena. Monitoring should be conducted throughout the year to determine DO levels at various seasons particularly through the late summer/fall. Circulation will increase in the harbour with winter storms as well as increased influence from the freshwater creeks which ultimately will bring DO levels back

up.

Bacteriological measurements during low flow levels seems to indicate minimal contamination is taking place in the harbour. Follow up surveys should be done to see what happens when flows increase at the STP. Will fecal coliform numbers increase with the excessive storm water entering the effluent?

There is indication of long term effects from Ladysmith STP effluent on the receiving environment. Sediment samples taken within the immediate vicinity of the outfall appear toxic. The toxicity of the sediment appears likely due to the build up of hydrogen sulphide. During sampling it was noted that the sediment was black within several centimeters of the surface and emitted a strong hydrogen sulfide odour. The Macoma bioassay had no living clams after the test period which is also a strong indication of the toxicity present in the sediment. The sediment deposition, however, may be attributable to historic sludge disposal procedures (i.e. through outfall). Further sediment analysis of parameters such as metals is recommended. Furthermore in monitoring the sediment, samples should be taken in a gradient beginning at the outfall and moving outward. This would indicate the level of toxicity and at what point the sediment becomes non-toxic. Perhaps divers could be used to inspect more restricted areas. Monitoring should be continued so as to find and record any changes that may occur.

While the results presented here are somewhat preliminary in nature, they are nevertheless of concern. Even though B.O.D. levels meet the present STP permit requirements, they may nevertheless remain high enough to impact the receiving environment, contributing to the apparent reduction of dissolved oxygen in the receivng environment and sediment toxicity through the build-up of hydrogen sulphide. The effluent continues to be toxic likely as a result of high B.O.D. levels and fluctuating ammonia levels. Infiltration continues to be a problem through the winter months leading to shellfish harvesting closures. Reducing present B.O.D. loadings, ammonia levels and dealing with the infiltration problem are the key to removing the shellfish closures and producing a non-toxic effluent. The two options presented for improving the Ladysmith sewage treatment plant should be further investigated.

REFERENCES

Anonymous. 1981a. Ladysmith Harbour Crown Foreshore Plan. British Columbia Ministry of Lands, Parks and Housing, 94pp.

Bourne, N., H. Rogers, H. Mahood and D. Neil, 1981. Water quality study of Ladysmith Harbour, B.C.. Canadian Technical Report of Fisheries and Aquatic Sciences No. 1026. Department of Fisheries and Oceans.

British Columbia Bioassay Task Force. 1982. Provincial guidelines and laboratory procedures for measuring acute lethal toxicity of liquid effluents to fish. 18 pp.

Clark, M.J.R. and W.P. Shera, 1985. Methods of aquatic data collection in B.C.; A catalogue. Ministry of Environment Manual No. 7, Ministry of Environment, Victoria, B.C.

Cooper, K.R. and B.H. Kay, 1975. Shellfish growing water sanitary survey of Ladysmith Harbour and outlying areas. Environmental Protection Service. Report No. EPS-5-PR-75-10.

Davis, J.C. 1975. Minimal dissolved oxygen requirements of aquatic life with emphasis on Canadian species: a review. J. Fish. Res. Board Can. 32: 2295-2332.

Dayton and Knight Ltd., 1991. Town of Ladysmith, Waste water treatment options for sewage treatment plant expansion. Vancouver, B.C.

Department of Fisheries and Oceans, 1988. Fish habitat inventory and information program. Stream summary catalogue, Subdistrict 17, South Coast Division, Fisheries Branch.

Kay, B.H. and D. Walker, 1985. Shellfish growing water bacteriological and sanitary survey of Ladysmith Harbour, Davis Lagoon, Boulder Point, and Sharpe Point to Yellow Point, B.C.. Environmental Protection Service, Pacific Region, Regional Program Report 84-18.

McDougal, I. and I. Boyd, 1984. 1982 water quality study of Ladysmith Harbour. Resource Quality Section, Water Management Branch, Province of B.C.

Nordin, R.N. and L.W. Pommen, 1986. Water quality criteria for nitrogen (nitrate, nitrite and ammonia). Ministry of Environment, Water Quality Unit, Victoria, B.C.

Patterson, J.H., 1975. An inventory of the fisheries resources of Ladysmith Harbour. Department of the Environment, Fisheries and Marine Service, Fisheries Operations South, Habitat Protection Unit.

Pommen, L.W., 1989. Approved and working criteria for water quality. Ministry of Environment, Water Management Branch, Resource Quality Section.

Pommen, L.W. and B. Holden, 1981. Ladysmith Harbour drogue tracking, October 6, 7 and 8, 1981. Aquatic Studies Branch, Ministry of Environment, Province of B.C. File 0322512-4.

Tevendale, T.J., 1973a. Shellfish growing water sanitary survey of Ladysmith Harbour, B.C., 1970. Environmental Protection Service. Report No. EPS 5-WP-73-1.

APPENDIX I

TABLES INDICATING WATER QUALITY DATA FOR LADYSMITH SEWAGE TREATMENT PLANT EFFLUENT FROM 1988 TO 1991

Table 1. The 1988 effluent data for the Ladysmith sewage treatment plant.

Date	N.F.R. (mg/L)	B.O.D. (mg/L)	Fecal Coliforms (CFU/100ml)
Jan 12	26	50	190000
Feb 11	28	54	1300
Mar 08	27	35	700
May 03	25	44	67000
June 13	34	92	440000
July 11	20	78	20000
Aug 16	34	78	3500
Sept 13	26	74	9500
Oct 25	28	140	150
Nov 15	18	25	250
Dec 13	22	35	90000

Note: N.F.R. = Nonfilterable residue (130 mg/L permit level)
B.O.D. = Biochemical oxygen demand (130 mg/L permit level)

Table 2. The 1989 effluent data for the Ladysmith sewage treatment plant.

Date	N.F.R. (mg/L)	B.O.D. (mg/L)	Fecal Coliforms (CFU/100ml)
Jan 10	23	46	650
Feb 07	29	56	90000
Mar 07	22	38	19000
Apr 11	36	57	800000
May 16	46	90	700
June 13	44	88	550
July 11	29	64	230000
Aug 15	44	102	3500
Sept 12	30	88	700
Oct 10	28	102	300
Nov 34	34	90	100
Dec 05	26	31	150

Note: N.F.R. = Nonfilterable residue
B.O.D. = Biochemical oxygen demand

Table 3. The 1990 effluent data for the Ladysmith sewage treatment plant.

Date	N.F.R. (mg/L)	B.O.D. (mg/L)	Fecal Coliforms (CFU/100ml)
Jan 16	20	40	35000
Feb 13	19	42	221000
Mar 13	18	34	11400
Apr 10	42	114	560
May 14	68	115	80
June 12	10	48	160
July 17	24	64	340
July 31	34	64	72000
Sept 25	36	136	190000
Oct 16	24	94	150
Nov 07	36	80	400000
Dec 11	23	29	60000

Note: N.F.R. = Nonfilterable residue (130 mg/L permit level)
B.O.D. = Biochemical oxygen demand (130mg/L permit level)

Table 4. The 1991 effluent data for the Ladysmith sewage treatment plant.

Date	N.F.R. (mg/L)	B.O.D. (mg/L)	Fecal Coliforms (CFU/100ml)
Jan 08	20	54	1600
Feb 12	24	43	12000
Mar 12	24	34	140
Apr 17	26	39	55000
May 14	38	89	500000
June 18	31	67	1500
July 16	30	67	2000000
Aug 22	27	83	800000
Sept 12	30	105	300000
Oct 08	26	91	4700
Nov 07	42	108	1000
Dec 03	22	44	35000

Note: N.F.R. = Nonfilterable residue
B.O.D. = Biochemical oxygen demand

APPENDIX II

SUMMARY OF WATER CHEMISTRY DATA FOR MONITORING SITES IN LADYSMITH HARBOUR

Ladysmith Harbour #1 Hydrolab Depth Profile

Date	Time	Depth	Temp	pH	Diss Oxy	ORP	Salinity
	hours	m	C	pH units	mg/L	mV	ppT
93/05/20	11:00	0	17.25	7.91	8.78	202	25.8
	11:00	2	16.15	8.03	10.03	199	26.2
	11:00	4	15.65	8.1	10.73	198	26.5
	11:00	5	13.63	8.11	9.12	199	27
93/07/06	8:40	0	17.99	7.92	8.26	195	25.1
	8:40	1	18.01	7.93	8.37	193	25.1
	8:40	2	17.84	7.95	8.29	190	25.2
	8:40	3	16.98	8	8.61	187	25.4
	8:40	4	16.96	7.99	7.9	187	25.4
	8:40	5	16.1	7.92	7.61	188	25.6
	8:40	6	15.14	7.87	6.34	190	25.9
	8:40	6.5	14.89	7.71	4.91	193	26

Ladysmith Harbour #2 Hydrolab Depth Profile

Date	Time	Depth	Temp	pH	Diss Oxy	ORP	Salinity
	hours	m	C	pH units	mg/L	mV	ppT
93/05/20	11:20	0	16.58	7.87	9.49	167	26
	11:20	2	15.83	8.01	10.54	166	26.5
	11:20	4	15.26	8.05	10.91	167	26.6
	11:20	6	13.49	8.09	10.81	169	27.1
93/07/06	9:20	0	18.13	7.85	8.84	170	25.2
	9:20	1	17.95	7.92	8.73	167	25.2
	9:20	2	17.46	7.91	8.19	166	25.3
	9:20	3	17.2	7.93	8.21	166	25.3
	9:20	4	16.72	7.95	8.13	165	25.5
	9:20	5	15.8	7.93	7.8	166	25.7
	9:20	6	15.31	7.86	6.84	169	25.9
	9:20	7	13.81	7.67	4.85	175	26.4
	9:20	8	13.08	7.63	4.66	175	26.6

Ladysmith Harbour #3 Hydrolab Depth Profile

Date	Time	Depth	Temp	pH	Diss Oxy	ORP	Salinity
	hours	m	C	pH units	mg/L	mV	ppT
93/05/20	11:35	0	15.59	8.01	11.75	160	26.5
	11:35	2	15.45	8.07	11.4	161	26.5
	11:35	4	15	8.06	10.91	162	26.6
	11:35	6	14.49	8.11	11.24	162	26.8
	11:35	8	12.4	8.12	11.13	165	27.3
	11:35	9	12.05	8.11	10.66	167	27.4
93/07/06	9:45	0	17.93	7.9	8.64	163	25.1
	9:45	2	17.6	7.97	8.61	161	25.2
	9:45	4	15.91	7.86	6.9	165	25.8
	9:45	6	14.21	7.72	5.81	169	26.2
	9:45	8	13.46	7.67	5.15	171	26.5
	9:45	10	12.27	7.64	5.18	172	26.9

Ladysmith Harbour #4 Hydrolab Depth Profile

Date	Time	Depth	Temp	pH	Diss Oxy	ORP	Salinity
	hours	m	C	pH units	mg/L	mV	ppT
93/05/20	11:50	0	15.55	8.11	12.84	153	26.7
	11:50	2	15.43	8.15	11.96	153	26.8
	11:50	4	15.25	8.15	11.77	154	26.8
	11:50	6	13.73	8.12	11.64	157	27
	11:50	8	11.36	8.11	11.6	160	27.6
	11:50	10	10.63	8.05	10.94	164	27.7
	11:50	12	10.14	8	10.48	166	27.8
	11:50	13	9.94	7.94	9.6	169	27.8
93/07/06	10:05	0	17.37	8	9.92	159	25.3
	10:05	2	16.99	8.11	9.86	153	25.2
	10:05	4	16.43	8.1	9.76	152	25.4
	10:05	6	15.43	8.17	10.1	151	25.2
	10:05	8	14.73	8.11	9.41	153	25.7
	10:05	10	14.63	8.1	9.24	153	25.6
	10:05	12	14.21	8.02	8.46	156	26
	10:05	14	14.34	7.91	7.22	159	26.2
	10:05	16	13.69	7.86	6.54	161	26.4
	10:05	17	12.39		5.73	166	26.9

Ladysmith Harbour #5 Hydrolab Depth Profile

Date	Time	Depth	Temp	pH	Diss Oxy	ORP	Salinity
	hours	m	C	pH units	mg/L	mV	ppT
93/05/20	12:05	0	15.53	8.25	12.95	141	26.9
	12:05	2	15.33	8.29	13.12	141	26.8
	12:05	4	15.12	8.3	13.17	142	26.9
	12:05	6	13.57	8.15	11.87	148	27
	12:05	8	11.54	8.1	11.54	152	27.5
	12:05	10	10.93	8.1	11.64	154	27.7
	12:05	12	10.62	8.1	11.72	155	27.8
	12:05	14	10.22	8.05	11.1	158	27.9
	12:05	16	10.04	7.94	9.46	164	27.9
	12:05	18	9.84	7.9	9.07	166	28.1
	12:05	19	9.65	7.86	8.48	157	28.1
93/07/06	10:30	0	17.29	8.02	9.9	147	25.3
	10:30	2	16.62	8.11	10.05	145	25.2
	10:30	4	16.23	8.16	10.12	144	25.2
	10:30	6	15.74	8.18	10.12	143	25.2
	10:30	8	15.1	8.15	9.85	144	25.4
	10:30	10	14.87	8.08	9.14	146	25.7
	10:30	12	13.91	8.04	8.79	148	25.9
	10:30	14	13.59	7.86	7.11	153	26.4
	10:30	16	12.51	7.75	6	158	26.8
	10:30	18	12.34	7.73	5.99	159	26.9
	10:30	20	11.27	7.73	5.83	160	27.4
	10:30	22	10.81	7.67	5.32	162	28

Ladysmith Harbour #6 Hydrolab Depth Profile

Date	Time	Depth	Temp	pH	Diss Oxy	ORP	Salinity
	hours	m	C	pH units	mg/L	mV	ppT
93/05/20	12:55	0	15.89	8.22	12.95	121	26.8
	12:55	2	15.08	8.29	13.52	122	27
	12:55	4	14.39	8.28	13.28	125	27.1
	12:55	6	12.37	8.27	13.86	127	27.5
	12:55	8	11.42	8.26	14	130	27.5
	12:55	10	10.78	8.19	13.07	134	27.7
	12:55	12	10.44	8.13	12.36	137	27.8
	12:55	14	10.31	8.11	11.86	139	27.8
	12:55	16	10.07	8.04	10.72	143	27.9
	12:55	18	9.92	8.02	10.43	146	28
	12:55	20	9.96	7.96	9.65	149	28.2
	12:55	25	9.29	7.83	8.01	156	28.3
	12:55	30	8.93	7.75	7.02	160	28.6
	12:55	35	8.75	7.79	7.48	162	28.7
	12:55	40	8.52	7.75	7.08	164	28.9
93/07/06	10:55	0	16.25	8.09	10.44	119	25.1
	10:55	2	16.06	8.13	10.34	118	25.1
	10:55	4	15.81	8.14	10.23	118	25.1
	10:55	6	14.52	8.09	9.8	121	25.4
	10:55	8	14.25	8.08	9.42	123	25.7
	10:55	10	12.91	7.86	7.44	131	26.4
	10:55	12	12.43	7.81	6.94	134	26.6
	10:55	14	12.18	7.83	7.04	135	26.8
	10:55	16	11.6	7.78	6.62	139	27.3
	10:55	18	11.46	7.7	5.74	142	27.4
	10:55	20	11.17	7.68	5.56	145	27.6
	10:55	25	10.78	7.75	6.12	144	28.2
	10:55	30	10.4	7.73	5.96	146	28.4
	10:55	35	10.23	7.73	5.85	147	28.5
	10:55	40	10.07	7.71	5.63	150	28.8
	10:55	45	9.99	7.7	5.4	151	28.4
	10:55	50	9.88	7.65	4.81	148	28.6



Intertidal mud flats at the head of Ladysmith Harbour.



Log boom situated on the west coast of inner harbour.



Ladysmith STP outfall located in the outer harbour.



View from Slag Point separating inner/outer harbour.